The Black–Scholes Formula via the Martingale Approach

Quant Finance Portfolio — Phase 1, Algorithm 1

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Abstract

We derive the Black–Scholes price of a European call option using the equivalent martingale measure framework. The argument is self-contained modulo the standard tools of stochastic calculus—Itô's lemma, Girsanov's theorem, and the fundamental theorems of asset pricing—which we state without proof and apply directly. Beyond the formula itself, we record three conceptual observations whose appreciation goes beyond the algebraic manipulation: the distinction between physical and risk-neutral probabilities of exercise, the change-of-numéraire interpretation of $N(d_1)$, and the way the formula encodes its own replicating portfolio.

1 Model

Fix a finite horizon $T > 0$ and a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\in[0,T]}, \mathbb{P})$ supporting a standard Brownian motion W, with $\{\mathcal{F}_t\}$ the $\mathbb{P}$-augmentation of the natural filtration of W. The market consists of two assets:

- A *risky* asset (the stock), whose price S follows a geometric Brownian motion under $\mathbb{P}$:

$$\mathrm{d}S_t = \mu S_t \,\mathrm{d}t + \sigma S_t \,\mathrm{d}W_t, \qquad S_0 > 0, \tag{1}$$

  with constants $\mu \in \mathbb{R}$ and $\sigma > 0$, paying no dividends.

- A *riskless* asset (the money-market account) with price $B_t = e^{rt}$, $r \geq 0$.

We seek the time-t price C_t of a European call option with strike $K > 0$ and maturity T, whose payoff at expiry is $(S_T - K)^+$.

2 Change of Measure

Define the discounted stock price $\tilde{S}_t \overset{\text{def}}{=} e^{-rt} S_t$. Applying Itô's lemma to (1),

$$\mathrm{d}\tilde{S}_t = (\mu - r)\tilde{S}_t \,\mathrm{d}t + \sigma \tilde{S}_t \,\mathrm{d}W_t. \tag{2}$$

For the model to admit no arbitrage, the discounted price must be a martingale under the pricing measure. The drift in (2) prevents this under $\mathbb{P}$, so we look for an equivalent measure $\mathbb{Q} \sim \mathbb{P}$ under which the drift vanishes.

Set the *market price of risk*

$$\lambda \overset{\text{def}}{=} \frac{\mu - r}{\sigma},$$

and define the Radon–Nikodym derivative

$$\left.\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}\right|_{\mathcal{F}_T} = \exp\!\left(-\lambda W_T - \tfrac{1}{2}\lambda^2 T\right).$$

Since λ is constant, Novikov's condition $\mathbb{E}^{\mathbb{P}}\left[\exp\left(\frac{1}{2}\int_0^T \lambda^2 \,\mathrm{d}s\right)\right] < \infty$ holds trivially. By Girsanov's theorem,

$$\widetilde{W}_t \stackrel{\text{def}}{=} W_t + \lambda t$$

is a standard Brownian motion under $\mathbb{Q}$. Substituting in (1) and (2),

$$\mathrm{d}S_t = rS_t \,\mathrm{d}t + \sigma S_t \,\mathrm{d}\widetilde{W}_t, \tag{3}$$

$$\mathrm{d}\tilde{S}_t = \sigma \tilde{S}_t \,\mathrm{d}\widetilde{W}_t. \tag{4}$$

Under $\mathbb{Q}$, the discounted price is a (local, in fact true) martingale, and the model is therefore arbitrage-free. Moreover, since the market has a single source of randomness driving a single risky asset, the second fundamental theorem of asset pricing implies that $\mathbb{Q}$ is the *unique* equivalent martingale measure, and the market is complete.

3 The Pricing Formula

By the first fundamental theorem of asset pricing, the no-arbitrage price at time t of any $\mathcal{F}_T$-measurable, sufficiently integrable contingent claim X is

$$\pi_t(X) = B_t \,\mathbb{E}^{\mathbb{Q}}\left[\frac{X}{B_T} \,\middle|\, \mathcal{F}_t\right].$$

For the European call,

$$C_t = e^{-r\tau}\mathbb{E}^{\mathbb{Q}}\left[(S_T - K)^+ \,\middle|\, \mathcal{F}_t\right], \qquad \tau \stackrel{\text{def}}{=} T - t. \tag{5}$$

4 The Distribution of S_T under $\mathbb{Q}$

To evaluate (5) we need the law of S_T under $\mathbb{Q}$, conditional on $\mathcal{F}_t$. Applying Itô's lemma to $\log S_t$ using (3),

$$\mathrm{d}(\log S_t) = \left(r - \tfrac{1}{2}\sigma^2\right)\mathrm{d}t + \sigma \,\mathrm{d}\widetilde{W}_t.$$

Integrating from t to T,

$$\log\left(\frac{S_T}{S_t}\right) = \left(r - \tfrac{1}{2}\sigma^2\right)\tau + \sigma\left(\widetilde{W}_T - \widetilde{W}_t\right),$$

and since $\widetilde{W}_T - \widetilde{W}_t \sim \mathcal{N}(0,\tau)$ under $\mathbb{Q}$ and is independent of $\mathcal{F}_t$,

$$S_T \,\middle|\, \mathcal{F}_t \stackrel{\mathbb{Q}}{=} S_t \exp\left(\left(r - \tfrac{1}{2}\sigma^2\right)\tau + \sigma\sqrt{\tau}\, Z\right), \qquad Z \sim \mathcal{N}(0,1). \tag{6}$$

5 Evaluating the Expectation

Let φ and N denote the standard normal density and CDF respectively. Substituting (6) into (5),

$$C_t = e^{-r\tau}\int_{\mathbb{R}} \left(S_t \exp\left((r - \tfrac{1}{2}\sigma^2)\tau + \sigma\sqrt{\tau}\, z\right) - K\right)^+ \varphi(z)\,\mathrm{d}z.$$

The integrand is positive iff

$$S_t \exp\left((r - \tfrac{1}{2}\sigma^2)\tau + \sigma\sqrt{\tau}\, z\right) > K \iff z > -d_2,$$

where

$$d_2 \stackrel{\text{def}}{=} \frac{\log(S_t/K) + (r - \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}. \tag{7}$$

Splitting the integral over the region where the payoff is positive,

$$C_t = \underbrace{e^{-r\tau} S_t \int_{-d_2}^{\infty} e^{(r-\sigma^2/2)\tau+\sigma\sqrt{\tau}\,z} \varphi(z)\,\mathrm{d}z}_{I_1} \;-\; \underbrace{e^{-r\tau} K \int_{-d_2}^{\infty} \varphi(z)\,\mathrm{d}z}_{I_2}. \tag{8}$$

Second integral. By symmetry of the standard normal density,

$$I_2 = e^{-r\tau} K\, \mathbb{P}(Z > -d_2) = e^{-r\tau} K\, N(d_2).$$

First integral. Combine the exponentials in the integrand and complete the square:

$$-\frac{z^2}{2} + \sigma\sqrt{\tau}\, z = -\frac{(z-\sigma\sqrt{\tau})^2}{2} + \frac{\sigma^2\tau}{2}.$$

The term $+\sigma^2\tau/2$ cancels the $-\sigma^2\tau/2$ inside $(r-\sigma^2/2)\tau$, and the prefactor $e^{r\tau}$ cancels the discount $e^{-r\tau}$. With the change of variable $u \overset{\text{def}}{=} z - \sigma\sqrt{\tau}$,

$$I_1 = S_t \int_{-d_2-\sigma\sqrt{\tau}}^{\infty} \varphi(u)\,\mathrm{d}u = S_t\, N\big(d_2 + \sigma\sqrt{\tau}\big) \overset{\text{def}}{=} S_t\, N(d_1),$$

where

$$d_1 \overset{\text{def}}{=} d_2 + \sigma\sqrt{\tau} = \frac{\log(S_t/K) + (r + \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}. \tag{9}$$

6 Result

Combining the two integrals:

Theorem 6.1 (Black–Scholes formula)**.** *The no-arbitrage price at time t of a European call option with strike K and maturity T on a non-dividend-paying stock following* (1) *is*

$$\boxed{C_t = S_t\, N(d_1) \;-\; K e^{-r\tau} N(d_2)}, \tag{10}$$

where $\tau = T - t$ and d_1, d_2 are given by (9)–(7)*.*

The price of the corresponding European put follows from put–call parity, an identity independent of any model assumption beyond no-arbitrage, which we record for completeness:

$$C_t - P_t = S_t - K e^{-r\tau}.$$

Combined with (10) and the symmetry $N(x) + N(-x) = 1$, this gives

$$P_t = K e^{-r\tau} N(-d_2) - S_t N(-d_1). \tag{11}$$

7 Conceptual Remarks

Remark 7.1 ($N(d_2)$ is not the physical probability of exercise)*.* By construction, $N(d_2) = \mathbb{Q}(S_T > K \mid \mathcal{F}_t)$, the *risk-neutral* probability that the option finishes in the money. Under the physical measure, the same probability would replace r by μ in the numerator of d_2; the gap between the two is, exactly, the market price of risk λ. Conflating these two probabilities is one of the most common conceptual errors in introductions to derivatives pricing.

Remark 7.2 ($N(d_1)$ and the change of numéraire). A second probabilistic interpretation, less commonly emphasised, is that

$$N(d_1) = \mathbb{P}^S(S_T > K \mid \mathcal{F}_t),$$

where $\mathbb{P}^S$ is the equivalent measure obtained by taking the stock S (rather than the bond B) as numéraire. Under $\mathbb{P}^S$, the process B_t/S_t is a martingale, and the same change-of-measure machinery used to construct $\mathbb{Q}$ applies. The simultaneous appearance of $N(d_1)$ and $N(d_2)$ in the formula is therefore not coincidental: it is the signature of two distinct numéraires being used implicitly. This perspective generalises far beyond Black–Scholes and is the natural setting for pricing exchange options, LIBOR-based interest-rate products, and most exotic derivatives.

Remark 7.3 (The formula encodes its own hedge). The decomposition $C_t = S_t N(d_1) - Ke^{-r\tau}N(d_2)$ is, literally, the time-t value of the replicating portfolio: long $N(d_1)$ shares, financed by borrowing $Ke^{-r\tau}N(d_2)$ at the riskless rate. When in the next algorithm we differentiate (10) with respect to S and obtain $\partial C/\partial S = N(d_1)$ (the *delta*), with all the φ-terms cancelling cleanly, this will be a manifestation of the same internal consistency: the formula is structured around its own hedge.